

FINAL REPORT

1.INTRODUCTION:

1.1 Project Overview:

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites is a data visualization project that analyses UNESCO World Heritage Sites across different countries and regions. The project focuses on understanding the distribution of heritage sites based on category (Cultural, Natural, Mixed), geographical location, and country-wise comparison.

This project is mainly useful for researchers, students, travellers, and history enthusiasts who want to explore global heritage data in an interactive and visual way. The analysis was performed using **Tableau** for creating dashboards and visualizations, and structured datasets were used to generate meaningful insights.

1.2 Purpose:

The main purpose of this project is to simplify and visualize complex heritage site data so that users can easily understand global patterns and trends. Many people find it difficult to compare heritage sites between countries or analyse category distribution using raw data.

This project solves that problem by presenting the information through interactive charts, maps, and dashboards. It helps users quickly identify which countries have the highest number of heritage sites, how sites are distributed by category, and how heritage sites are spread across continents. This makes the data more accessible, informative, and engaging.

2.IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template:

Students, researchers, and travellers want to explore and understand information about **UNESCO World Heritage Sites**. However, the available information is scattered across multiple websites, mostly in long text formats, and lacks clear visual representation.

Because of this, users find it difficult to analyse country-wise distribution, compare regions, understand site categories (**cultural, natural, mixed**), and gain meaningful insights quickly.

Therefore, there is a need for an interactive, user-friendly platform that presents **UNESCO World Heritage** data in a clear, visual, and analytical format to help users easily explore and understand heritage information.

I'AM	A student, researcher, or traveler who wants clear and reliable information about UNESCO World Heritage Sites.
I'M TRYING TO	Explore, compare, and gain meaningful insights about different UNESCO World Heritage Sites easily.
BUT	The information is scattered across many websites, mostly in long paragraphs with very little visual representation.
BECAUSE	Text-heavy and unorganized content makes it difficult to analyze, compare, and understand quickly.
WHICH MAKES ME FEEL	Confused, overwhelmed, and frustrated while searching for useful insights.



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel	Problem Statement (PS)
PS-1	A student / data analyst	Compare UNESCO World Heritage Sites by country and category in Tableau	It takes too much time to analyse	The dataset is large and lacks interactive filters and clear visual comparisons	Frustrated	PS-1
PS-2	A researcher / presenter	Create meaningful insights from UNESCO World Heritage data	I am unable to identify patterns quickly	The raw data is not structured into dashboards, maps, or trend visuals	Overwhelmed and confused	PS-2

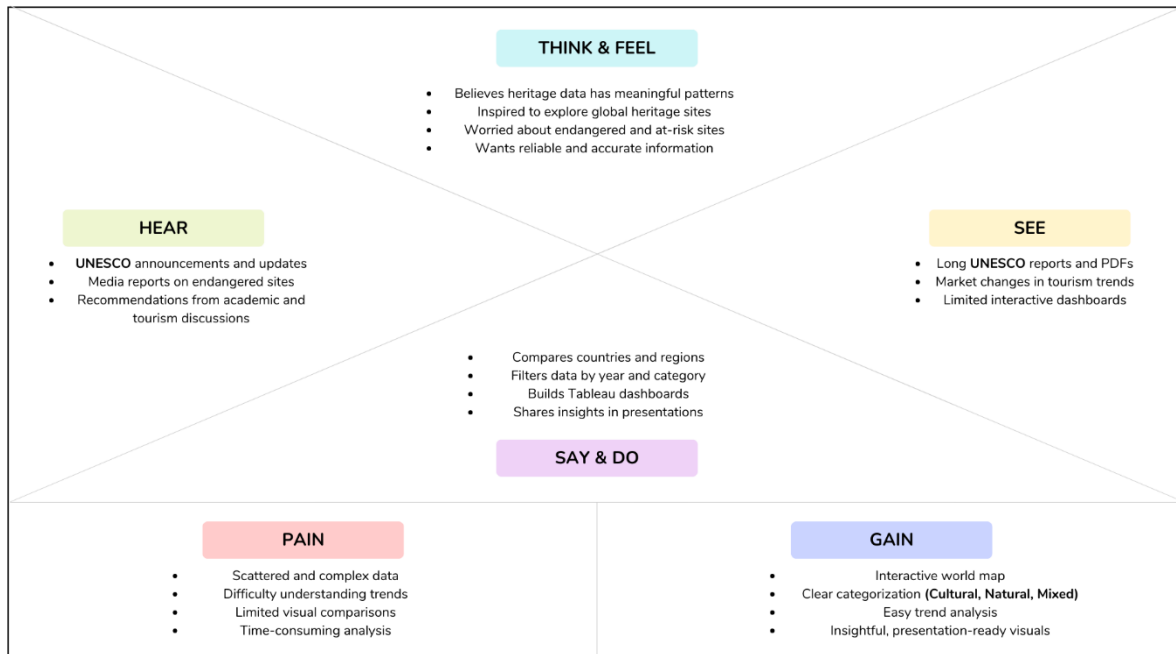
2.2 Empathy Map

An empathy map is a simple visual tool that helps us understand users' behaviours, needs, and challenges from their perspective.

For the Heritage Treasures – UNESCO World Heritage Sites Analysis in Tableau project, the empathy map helps identify what users think, feel, see, say, and struggle with when exploring heritage data. It highlights their pain points, such as scattered information and difficulty understanding trends, and their goals, such as interactive maps and clear visual insights.

By understanding the real problems users face, the project can deliver a more meaningful, user-centered Tableau dashboard that provides clear, insightful, and impactful analysis.

EMPATHY MAP



2.3 Brainstorming:

Brainstorming was conducted to generate creative ideas for designing and improving the Heritage Treasure – Cultural Site Analysis & Visualization System. The session encouraged open discussion, innovative thinking, and collaboration among all team members without criticism or judgment.

The primary objective of the brainstorming session was to identify features, improvements, and analytical insights that could enhance the effectiveness of the heritage data management and visualization system.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & Idea Prioritization

Use this template for your brainstorming sessions to analyse Union Budget allocations, trends, and their impact on India's sustainable growth.

- 30 minutes to prepare
- 1 hour to collaborate
- 3-8 people recommended

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Before you collaborate

A little bit of preparation goes a long way with session. Here's what you need to do to get going.

10 minutes

- Team gathering**
Check who's available - join in the session and send an invite. Share relevant materials if necessary.
- Set the goal**
Define the problem you'll be focusing on: evolving Union Budget allocations for sustainable growth.
- Learn how to use the facilitation tools**
Test your facilitation tools to ensure a smooth and productive session.

[Open articles](#)

Tip: Don't hold back! Focus on out-of-the-box ideas. It's okay if some ideas seem crazy - we'll refine later.

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

How might we better analyse Union Budget allocations for Sustainable growth?

Key rules of brainstorming

- Work in an open and positive session
- No judgment
- Go for volume
- Encourage wild ideas
- Listen to others
- Be visual

Step-2: Brainstorm, Idea Listing and Grouping



Brainstorm & Idea Prioritization

Write down any ideas related to UNESCO World Heritage Sites, their distribution, and significance.

20 minutes

Participant 1	Participant 2	Participant 3	Participant 4
Preservation Efforts			
Endangered Sites			
Tourism Impact			

Participant 5	Participant 6	Participant 7	Participant 8

Reference: <https://mural.com/templates/brainstorm-and-idea-prioritization>

Group Ideas

Take time refining your ideas while clustering similar or related notes as you. In this step, revisit and group the ideas from the previous step. Create clear categories based on theme, and see if you drill down into smaller sub-groups.

20 minutes

Cultural Preservation

UNESCO Funding
Heritage Education

Preservation Challenges

Endangered Sites

Sites at Risk
Legal Protection

Conservation Funding

Tourism & Impact

Sustainable Tourism
Visitor Management

Overcrowding

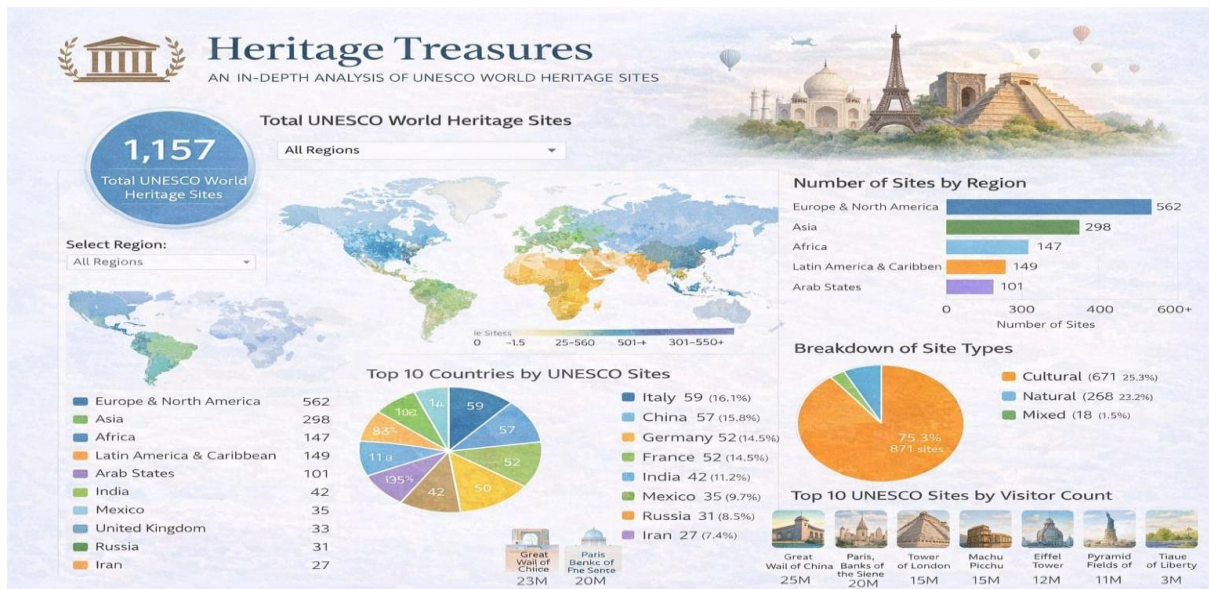
Educational & Outreach

Awareness Campaigns
Research Initiatives

Local Community Involvement
Initiatives

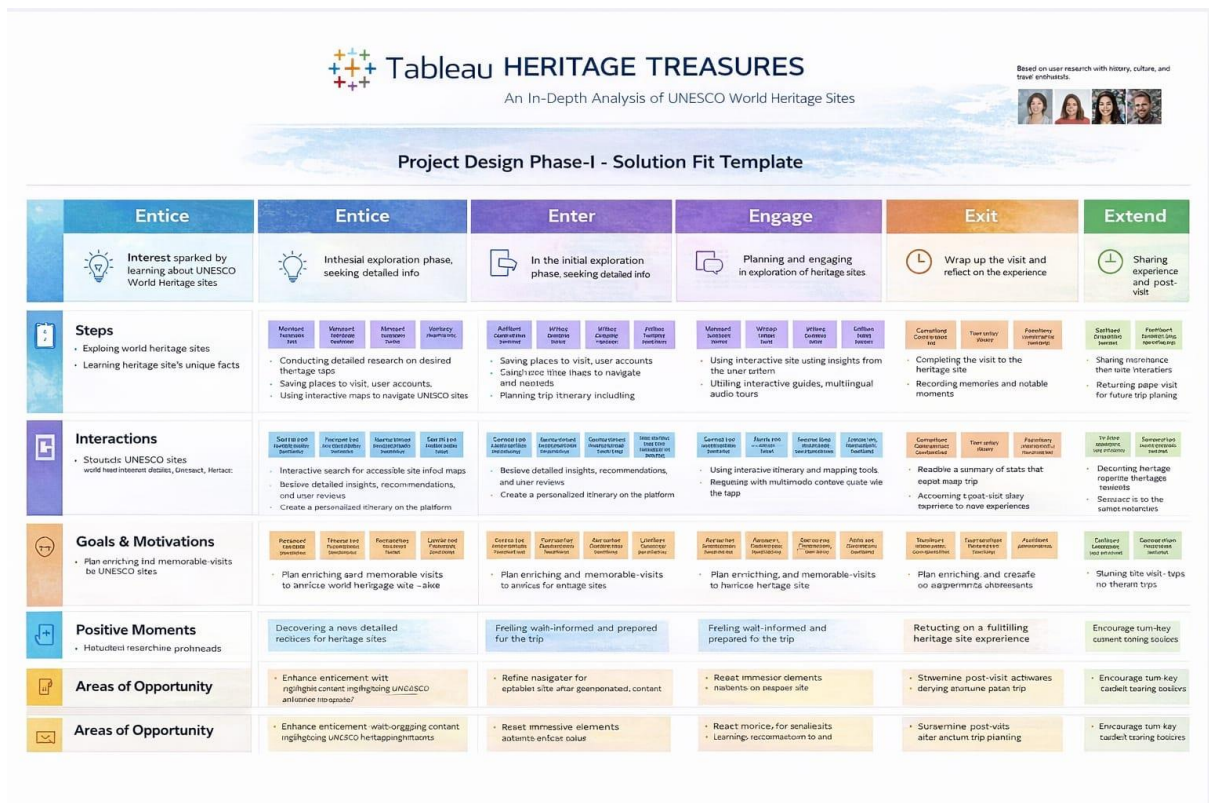
Tip: Group similar notes to better categorize your ideas. Revisit and refine for clarity.

Step-3: Idea Prioritization



3.REQUIREMENT ANALYSIS:

3.1 Customer Journey Map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction From Database	Downloading The Dataset
FR-2	Data Preparation	Explanation Video Links ● Data Loading ● Data Cleaning
FR-3	Data Visualizations	● No. Of Unique Visualizations ● Visualizations ○ Country Per Region ○ Top 10 regions by area ○ Regions by heritage ○ Top 10 Danger Sites prone to Extinction ○ Year forecasting of heritages ○ Categories by sites count ○ Sites count per region ○ Analysis of Danger sites and area of heritages
FR-4	Dashboard	● Dashboard1 ● Dashboard2
FR-5	Story	No. Of Scenes of Story
FR-6	Performance Testing	● Amount Of Data loaded ● Utilization Of Data Filters ● No Of Visualizations / Graphs
FR-7	Web Integration	Go to dashboard / story click on share button on the top ribbon

		Dashboard and story embed with UI with Flask
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Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

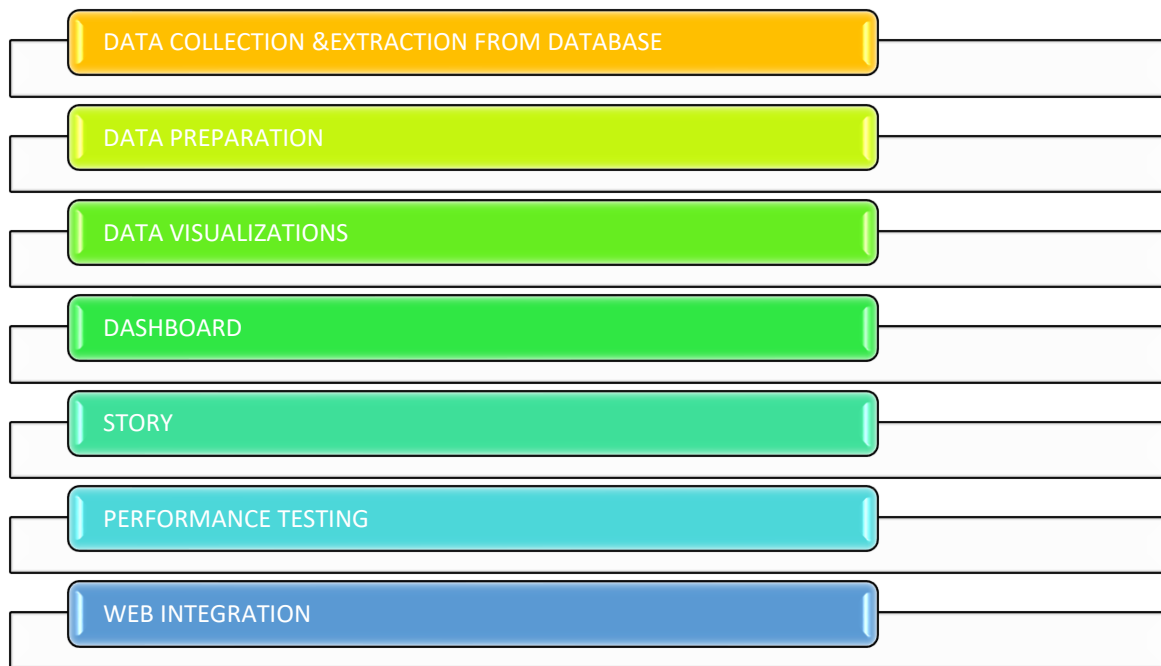
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The “ <i>Heritage Treasures</i> ” Tableau dashboard shall be simple, intuitive, and easy to navigate, enabling users to efficiently explore and analyse data on UNESCO World Heritage Sites . Clear visualizations, interactive filters, and a well-structured layout shall enhance understanding and support informed decision-making with minimal effort.
NFR-2	Security	Access controls, user authentication, and data privacy measures will be implemented to maintain data integrity and prevent misuse. The system shall ensure that only authorized users can modify or manage the dashboard, while general users can view the data safely and securely.
NFR-3	Reliability	The system shall be designed to minimize downtime, prevent data inconsistencies, and ensure continuous availability, enabling users to confidently rely on the dashboard for accurate analysis and informed decision-making.
NFR-4	Performance	The system shall provide fast and responsive performance when loading, filtering, and visualizing data in the “ <i>Heritage Treasures</i> ” Tableau dashboard. Data related to UNESCO World Heritage Sites shall be processed and displayed efficiently, ensuring minimal loading time and smooth interaction to support timely analysis and decision-making.
NFR-5	Availability	The system shall ensure high availability of the “ <i>Heritage Treasures</i> ” Tableau dashboard so users can access data on

		UNESCO World Heritage Sites whenever required. The dashboard shall remain accessible with minimal downtime, ensuring continuous access for analysis and informed decision-making
NFR-6	Scalability	The system shall be scalable to accommodate increasing volumes of data and a growing number of users in the <i>“Heritage Treasures”</i> Tableau dashboard. As more UNESCO World Heritage Sites are added or additional analytical features are introduced, the system shall maintain efficient performance without degradation.

3.3 Data flow Diagrams

The Data Flow Diagram (DFD) for the *“Heritage Treasures”* project shows how data about **UNESCO World Heritage Sites** flows through the system. The data is collected from the UNESCO dataset, processed and organized in Tableau, and then displayed as interactive dashboards.

Users apply filters such as country, region, and heritage type, and the system processes the input to generate updated charts and maps. The DFD clearly represents how data enters, is processed, stored, and presented for analysis.



3.4 Technology Stack:

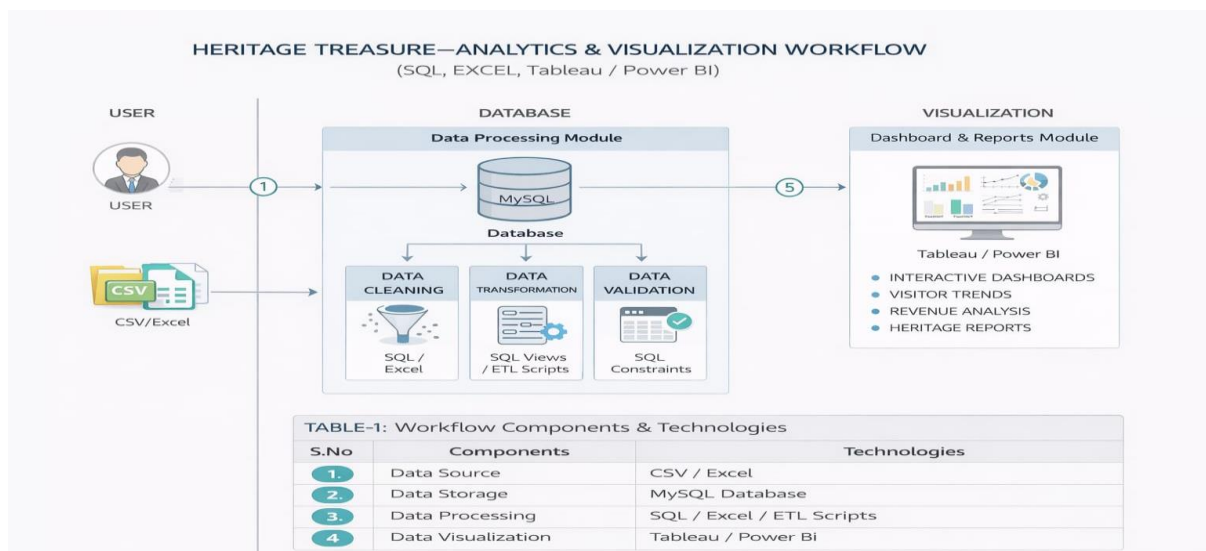


Table-1 : Components & Technologies:

S.n o	Component	Description	Technology

1	Data Source	Contains raw heritage site data such as visitor count, revenue, location, and category	CSV / Excel
2	Data Storage	Stores organized heritage data securely for analysis	MySQL
3	Data Visualization	Builds interactive dashboards, charts, and maps for analysis	Tableau / Power BI
4	Reporting	Presents insights like regional comparison, revenue trends, and visitor growth	Dashboard Reports
5	Dashboard Interaction	Allows users to explore data dynamically using filters and slicers	Tableau Filters / Power BI Slicers
6	Data Validation	Ensures accuracy, consistency, and reliability of data	Excel / SQL Constraints
7	Data Processing	Cleans data, removes errors, handles missing values, and creates KPIs	SQL / Excel
8	KPI Calculation	Computes total visitors, total revenue, growth %, and average visitors per site	SQL & Dashboard Calculated Fields

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
	Open-Source Frameworks	Used open-source tools for storing, processing, and visualizing heritage data.	MySQL, SQL, Excel, Tableau / Power BI
	Security Implementations	Implemented login authentication and restricted database access. Data validation applied.	RBAC, SQL Authentication, HTTPS

	Scalable Architecture	Followed 3-tier architecture (Database, Processing, Dashboard) for easy expansion.	MySQL, SQL, Tableau / Power BI
	Availability	Dashboard accessible online for users anytime.	Tableau Server / Power BI Service
	Performance	Optimized queries and used filters for fast dashboard performance.	SQL Indexing, Data Extracts

4. PROJECT DESIGN

4.1 Problem Solution Fit:

The Problem–Solution Fit in this project means identifying the difficulty faced by students and researchers in analyzing large UNESCO World Heritage datasets, and developing an interactive Tableau dashboard that effectively solves this problem through clear and structured visualizations. It helps understand user behavior patterns such as confusion with raw data, difficulty in comparison, and challenges in identifying trends, and ensures that the solution directly addresses these issues.

Purpose:

- Transform complex UNESCO heritage data into simple, interactive visual insights that support effective learning and research.
- Enable faster understanding and comparison of global heritage trends through structured dashboards.
- Enhance analytical thinking and storytelling by presenting data in a clear and engaging format.
- Reduce confusion caused by raw datasets and improve clarity in decision-making.
- Create a user-centered visualization system that makes heritage data accessible, insightful, and impactful for target groups

Template:

PROJECT DESIGN PHASE-I - SOLUTION FIT TEMPLATE

CUSTOMER SEGMENTS	CUSTOMER CONSTRAINTS	AVAILABLE SOLUTIONS	JOBS - TO - BE - DONE/PROBLEMS	PROBLEM ROUTE CAUSE
- Students (Geography / History learners) - Researchers & Data Analysts - Tourists & Travel Planners - Government / Policy Makers - Cultural Heritage Enthusiasts	- Customer Constraints - Limited technical knowledge - No proper visualization tools - Time constraints - Data overload - Lack of interactive filtering options	- UNESCO official website (raw data, less visual) - Research papers (complex data) - Static reports (not interactive) - Google search (unstructured information)	- Understand distribution of UNESCO sites by category (Cultural, Natural, Mixed) - Analyze endangered heritage sites over years - Compare countries based on number of heritage sites - Identify trends in sites under danger list - Study area coverage of endangered sites	- Heritage data is large and complex - No single interactive visualization platform - Limited awareness about endangered sites trends
	BEHAVIOUR		TRIGGERS	
	- Users: - Search heritage data online - Compare countries manually - Look for visual charts - Prefer dashboards over long reports		- Academic research requirement - Tourism planning - Environmental awareness programs - News about endangered heritage sites - Government policy discussions	
YOUR SOLUTION		CHANNELS OF BEHAVIOUR		
- Develop an Interactive Tableau Dashboard that: - Visualizes heritage site categories - Shows country-wise distribution - Displays endangered site trends over years - Includes filters for easy comparison - Makes data simple, clear, and interactive		- Educational websites - Research portals - Tableau dashboards - Travel blogs		

4.2 Proposed Solution:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
	Problem Statement (Problem to be solved)	The UNESCO World Heritage dataset is large and complex, making it difficult to analyze, compare countries and categories, and identify meaningful trends effectively.
	Idea / Solution description	Develop an interactive Tableau dashboard that converts complex UNESCO heritage data into clear visual insights, enabling easy comparison, trend analysis, and meaningful understanding.
	Novelty / Uniqueness	This project uniquely presents UNESCO World Heritage data through an interactive Tableau dashboard, enabling clear comparison, trend analysis, and engaging visual storytelling.
	Social Impact / Customer Satisfaction	This project improves awareness and understanding of UNESCO World Heritage Sites by making data accessible and easy to explore. It

		enhances user satisfaction by providing clear, interactive visual insights that support learning, research, and informed decision-making.
	Business Model (Revenue Model)	Revenue can be generated through premium dashboard access, institutional subscriptions, and partnerships with tourism or cultural organizations for advanced data insights.
	Scalability of the Solution	The solution is highly scalable as it can be expanded to include additional global datasets, real-time updates, and advanced analytics features. The dashboard can also be adapted for different users such as researchers, educational institutions, tourism boards, and policy makers.

4.3 Solution Architecture:

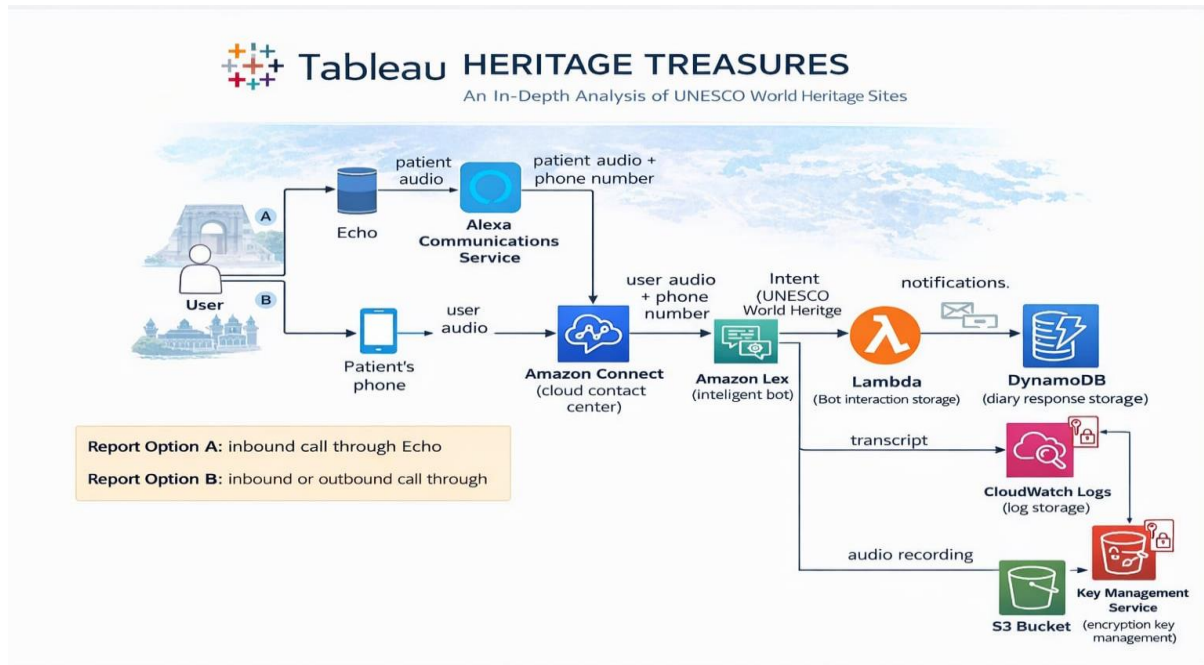
In this project, solution architecture bridges the gap between the challenge of analyzing large **UNESCO World Heritage** datasets and the use of Tableau as a visualization technology.

Its goals are to:

- Identify the most suitable data visualization approach to simplify and analyze **UNESCO** heritage data effectively.
- Define the structure of the interactive Tableau dashboard, including maps, comparison charts, filters, and trend analysis.
- Outline key features such as country-wise comparison, category classification, and year-wise growth tracking.
- Establish the data preparation process, including data cleaning, transformation, and integration into Tableau.

- Provide a clear framework for designing, developing, and presenting the dashboard in a structured and user-friendly manner.

Solution Architecture Diagram:



5. Project Planning And Scheduling:

5.1 Project Planning:

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection Module	HT-Us-1	As a system user, I can input heritage site details (name, location, category, year established).	3	High	Balasaraswathi seepana

Sprint-1	Data Collection Module	HT-Us-2	As a user, I can record visitor count and revenue details for each site.	5	High	Balasaraswathi seepana
Sprint-2	Site Classification	HT-Us-3	As a user, I can classify heritage sites by region and category (Temple, Fort, Museum).	4	Low	Biddika Umarani
Sprint-1	Visitor Analysis	HT-Us-4	As a user, I can calculate total visitors and average visitors per site.	3	Medium	Biddika Umaran
Sprint-1	Revenue Analysis	HT-Us-5	As a user, I can calculate total revenue and revenue per visitor.	5	High	Banda Sireesha
Sprint-2	Visualization Dashboard	HT-Us-6	As a user, I can view interactive graphs for visitor trends and revenue analysis.	8	High	Bandi Geethika
Sprint-3	Map Integration	HT-Us-7	As a user, I can view heritage site locations on an interactive map.	5	High	Bandi Geethika
Sprint-4	Final Dashboard	HT-Us-8	As a project admin, I can generate final heritage performance analysis reports.	4	High	Bandi Geethika

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	Data collection & extraction from the data base	6 Days	4 Feb 2026	10 Feb 2026	20	10 Feb 2026
Sprint-2	Data Preparation	3Days	10 Feb 2026	13 Feb 2026	20	13 Feb 2026
Sprint-3	Data Visualization	3 Days	13 Feb 2026	16 Feb 2026	20	16 Feb 2026
Sprint-4	Dashboard	2 Days	16 Feb 2026	18 Feb 2026	20	18 Feb 2026
Sprint-5	Story	2 Days	19Feb 2026	21Feb2026	20	21Feb2026
Sprint-6	Performance Testing	2 Days	21Feb 2026	23 Feb 2026	20	23 Feb 2026
Sprint-7	Web Integration	4 Days	23Feb 2026	27 Feb 2026	20	27 Feb 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

6.FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance testing

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S. No.	Parameter	Screenshot / Values
	Data Rendered	UNESCO World Heritage dataset successfully loaded and visualized in Tableau (Country, Category, Year Inscribed, Region fields used).
	Data Preprocessing	Data cleaned by removing null values, correcting country names, formatting date fields, and categorizing heritage types (Cultural, Natural, Mixed)
3.	Utilization of Filters	Interactive filters applied for Country, Region, Category, and Year to enable dynamic comparison and analyse
4.	Calculation fields Used	Calculated fields created for Total Sites per Country, Year-wise Growth, and Category Percentage Distribution.
5.	Dashboard design	No of Visualizations / Graphs - ● Packed bubble chart ● Map ● Filled Map ● Highlight table ● Line chart ● Bar chart ● Dual axis chart ● Pie chart
6	Story Design	No of Visualizations / Graphs - ● Countries per Region ● Top 10 regions by area

		- Regions by Heritage ended - Top 10 Danger Sites prone to Extinction - Year Forecasting of Heritages - Categories by sites count - Analysis of Danger sites and Area of heritages - Site Count per Region
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7.RESULTS:

VISUALIZATIONS:

7.1 : Countries per region



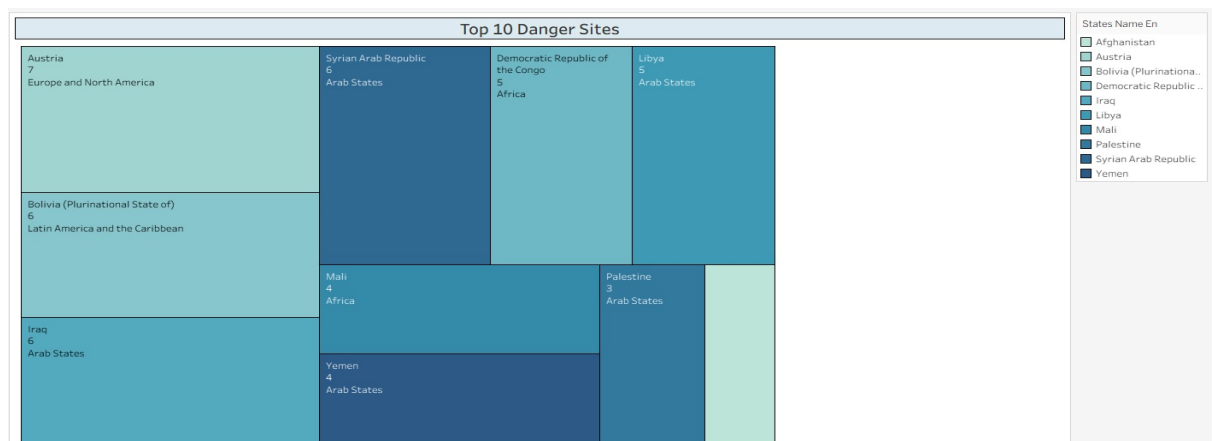
7.2 : Top 10 Regions by Area



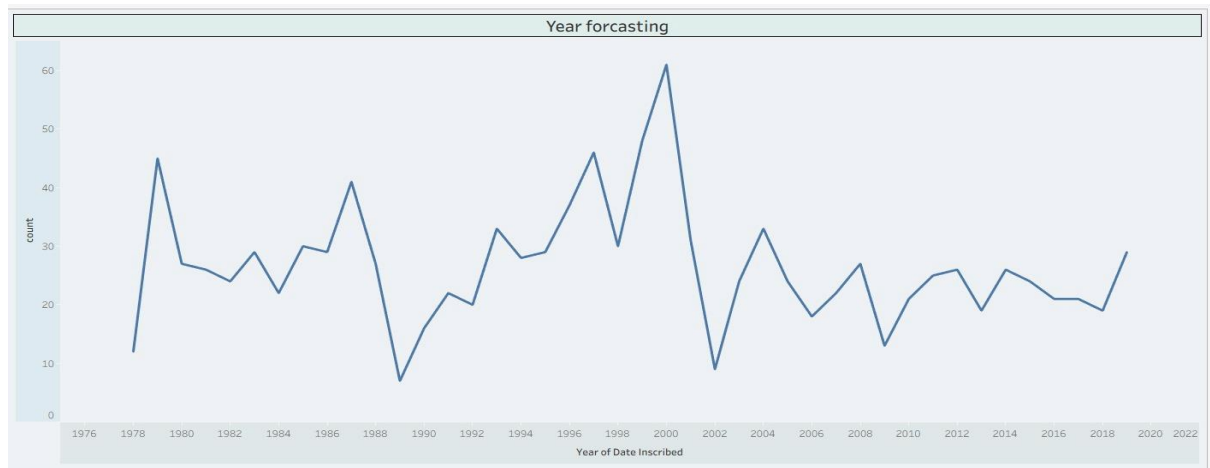
7.3 : Regions by heritages ended



7.4 : Top 10 Danger Sites Prone to Extinction



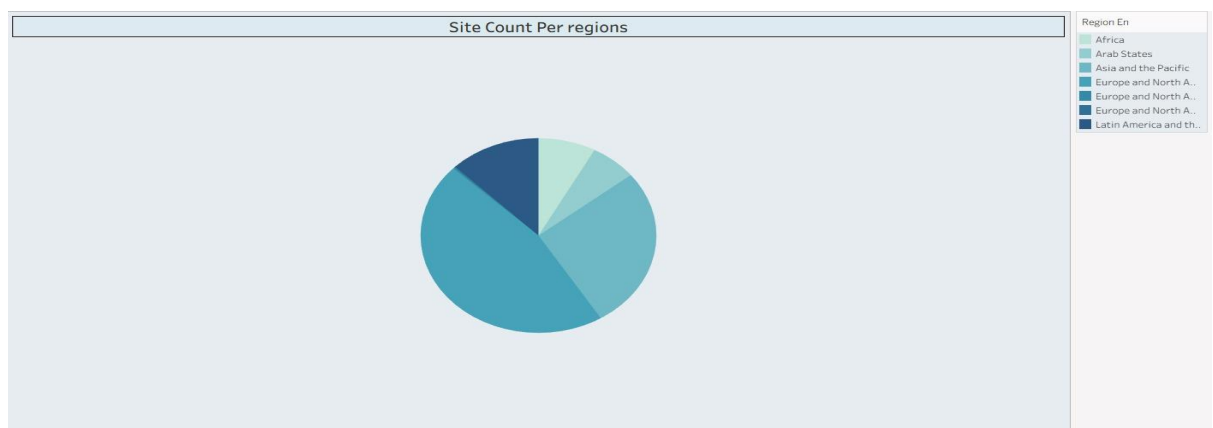
7.5 : Year Forecasting of Heritages



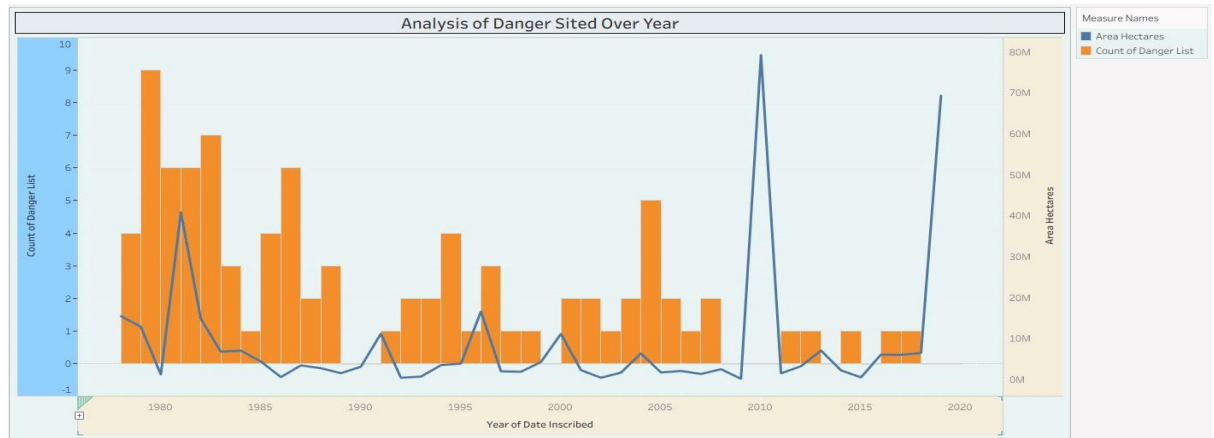
7.6 : Categories by Sites Counts



7.7 : Site Count Per Region

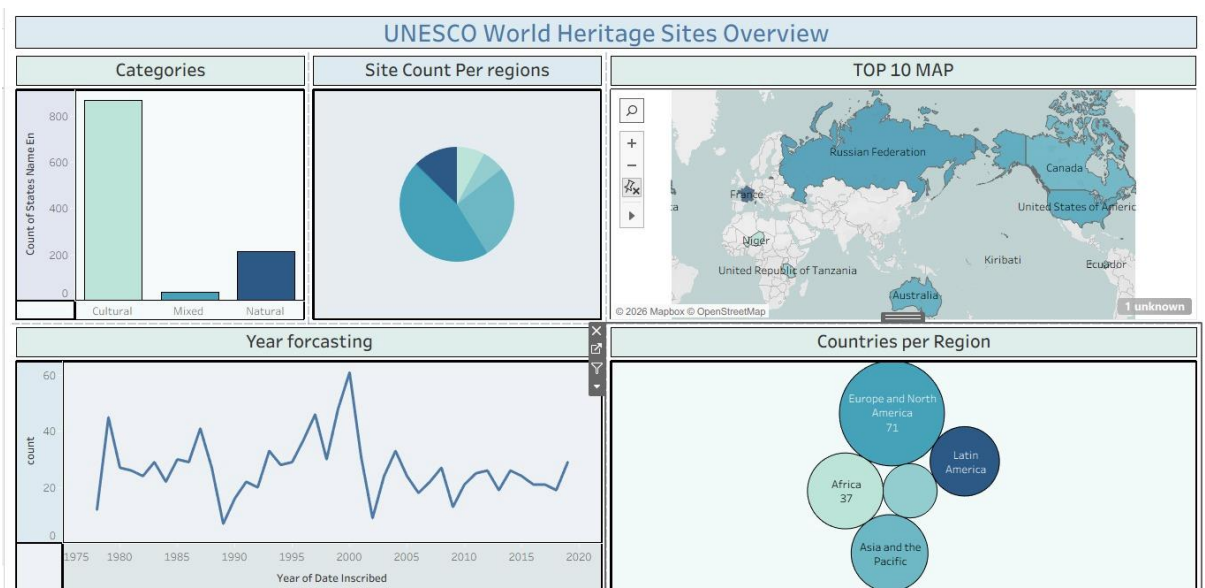


7.8 Analysis of Danger Sites and Area of Heritage



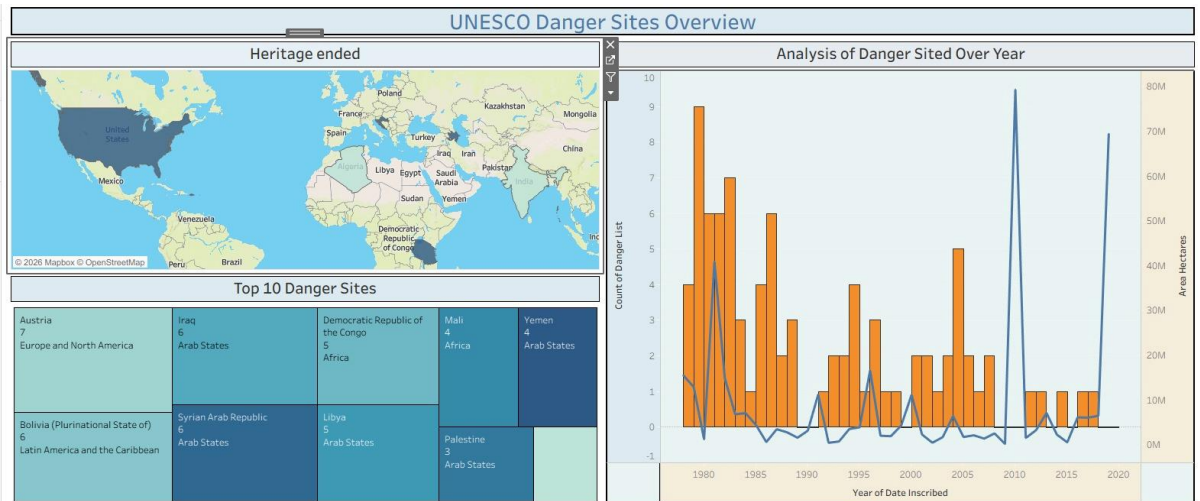
DASHBOARD1:

- UNESCO World Heritage Sites Overview



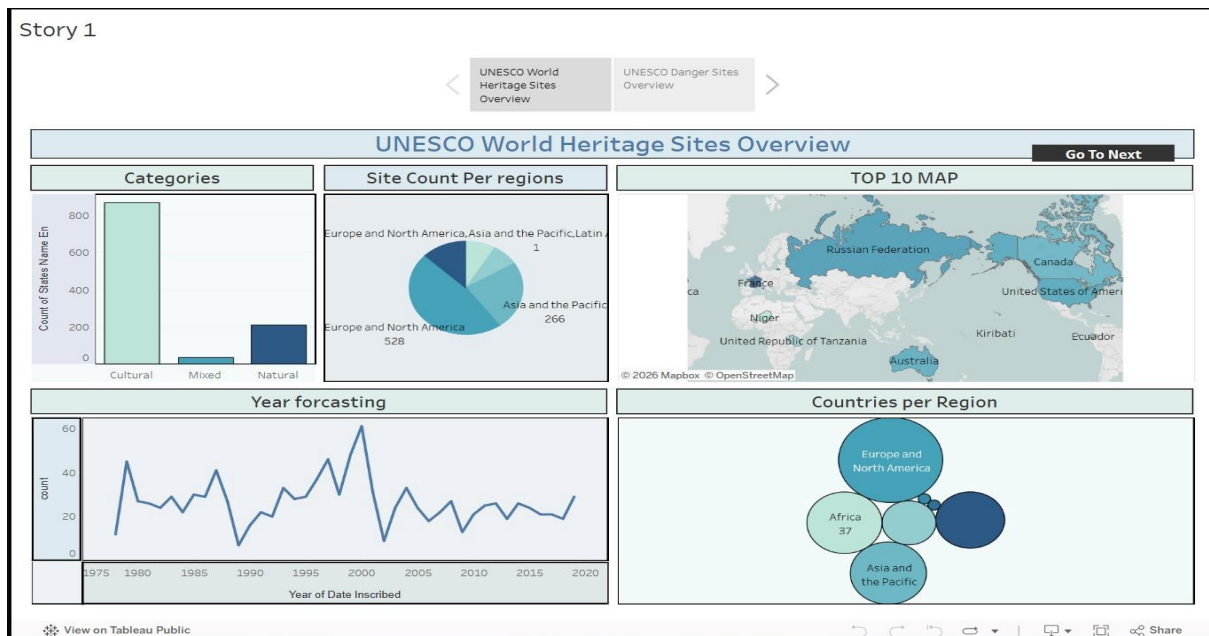
DASHBOARD2:

- UNESCO Danger sites Overview



STORY1

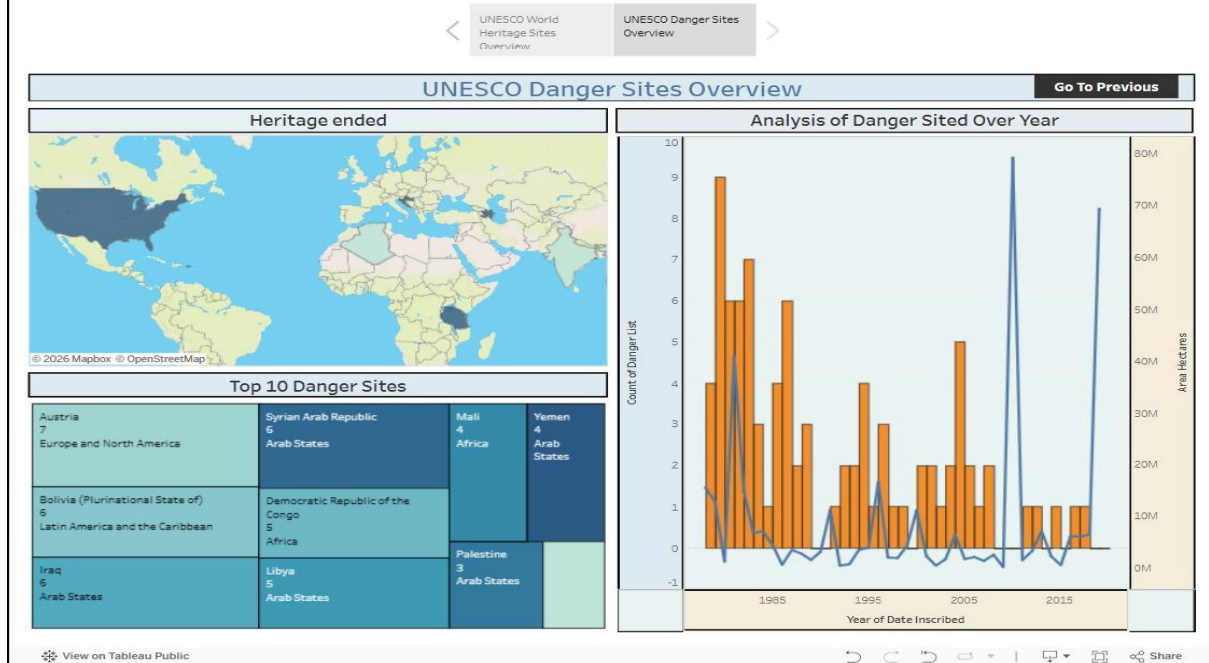
- UNESCO World Heritage Sites Overview



STORY2

- UNESCO Danger sites Overview

Story 1



WEB INTEGRATION:



8.ADVANTAGE AND DISADVANTAGES:

- Advantages

- **Clear Data Visualization:** The dashboard presents sales and performance data in charts and graphs, making complex information easy to understand.
- **Better Decision Making:** Helps identify trends, top-performing products, and high-profit regions for strategic planning.
- **Time Saving:** Reduces manual calculations by providing automated summaries and comparisons.
- **Interactive Features:** Filters and slicers allow users to analyse data by category, region, or date.
- **Organized Structure:** Data is well arranged in a structured format, improving readability.
- **Performance Tracking:** Enables monitoring of sales growth, profit margins, and overall business performance.
- **Disadvantages**
 - **Dataset Dependency:** Accuracy of insights depends completely on the quality and correctness of the dataset.
 - **Limited Real-Time Updates:** Data needs to be refreshed manually if not connected to a live source.
 - **Performance Issues:** Large datasets may slow down dashboard loading time.
 - **Technical Knowledge Required:** Basic understanding of Tableau is needed to update or modify the dashboard.
 - **Limited Without Integration:** Advanced analysis may require additional tools or API integration.

9. CONCLUSION:

In this project, I successfully designed and developed an interactive sales dashboard to analyse overall business performance in a clear and meaningful way. The dashboard presents important metrics such as total sales, profit, category-wise performance, and regional analysis using visual charts and filters. It enables users to quickly identify trends, compare different segments,

and evaluate business growth over time. The project demonstrates effective data cleaning, visualization techniques, and dashboard design skills. Overall, this dashboard transforms raw data into actionable insights, improving analytical ability and supporting smart, data-driven decision-making.

10. FUTURE SCOPE:

In the future, this project can be enhanced by integrating live API data to enable real-time updates and more accurate analysis. User login functionality can be added to provide personalized dashboards and secure access. A mobile-friendly version can be developed to improve accessibility across devices. Additionally, incorporating more datasets such as customer demographics and market trends can provide deeper insights and advanced analytical capabilities for better business decisions.

11. APPENDIX:

- **DATA SET LINK:**



- **SOURCE CODE**

https://drive.google.com/drive/folders/1oUAXyd8MP0DZvoXmg3qf7fOj6Vrhx2Sz?usp=drive_link

- **GIT HUB**

- [Geethika-Bandi/Heritage-treasures-An-in--depth-analysis-of-UNESCO-world-Heritage-sites-in-Tableau](https://github.com/Geethika-Bandi/Heritage-treasures-An-in--depth-analysis-of-UNESCO-world-Heritage-sites-in-Tableau) <https://github.com/Geethika-Bandi/Heritage-treasures-An-in--depth-analysis-of-UNESCO-world-Heritage-sites-in-Tableau>

- **PROJECT DEMO LINK**

<https://photos.app.goo.gl/UienSUX3vBAMRipv7>